

Spanner 和 BigQuery：即時詐欺防禦盾

來源：<https://codelabs.developers.google.com/spanner-bigquery-graph?hl=zh-tw>

步驟數：8

瞭解 Spanner 和 BigQuery 如何為遊戲建立即時詐欺防禦機制。

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1. 簡介

歡迎來到《**Petverse**》後端，這是一款多人線上遊戲，玩家可以建立動物虛擬化身、探索世界，以及交易遊戲代幣。

最近，遊戲經濟面臨威脅。有錢的狗狗玩家帳戶中，大筆金額全數轉換成頂級鮭魚股票，我們懷疑有一群「羅賓漢貓」從富有的獵犬身上偷取食物，餵食大量貓咪。

在本程式碼研究室中，您將建構**即時詐欺防禦盾**，以偵測首腦和自動機器人集團。您將體驗 **Spanner** 中的作業資料如何與 **BigQuery** 中的分析資料匯聚，並運用圖形 (GQL) 和關聯查詢偵測及調查複雜的詐欺模式。

學習內容

- 使用 **BigQuery 持續查詢**，在即時串流中偵測異常狀況，並將結果推送至 Spanner。
- 使用 **Spanner Graph** 追蹤財務關係，調查詐欺集團。
- 使用 **Spanner 向量搜尋** 找出相似的個人資料說明，找出機器人帳戶。
- 使用 **BigQuery Graph** 分析大量通訊記錄，找出協調圖。

軟硬體需求

- 已啟用計費功能的 Google Cloud 專案。
- 具備 SQL、終端機指令和 Python 的基本知識。
- 您可能需要 GitHub 帳戶 (程式碼代管於 GitHub)。

目標對象：中階開發人員、資料工程師和架構師。

預計總時長：45 到 60 分鐘。

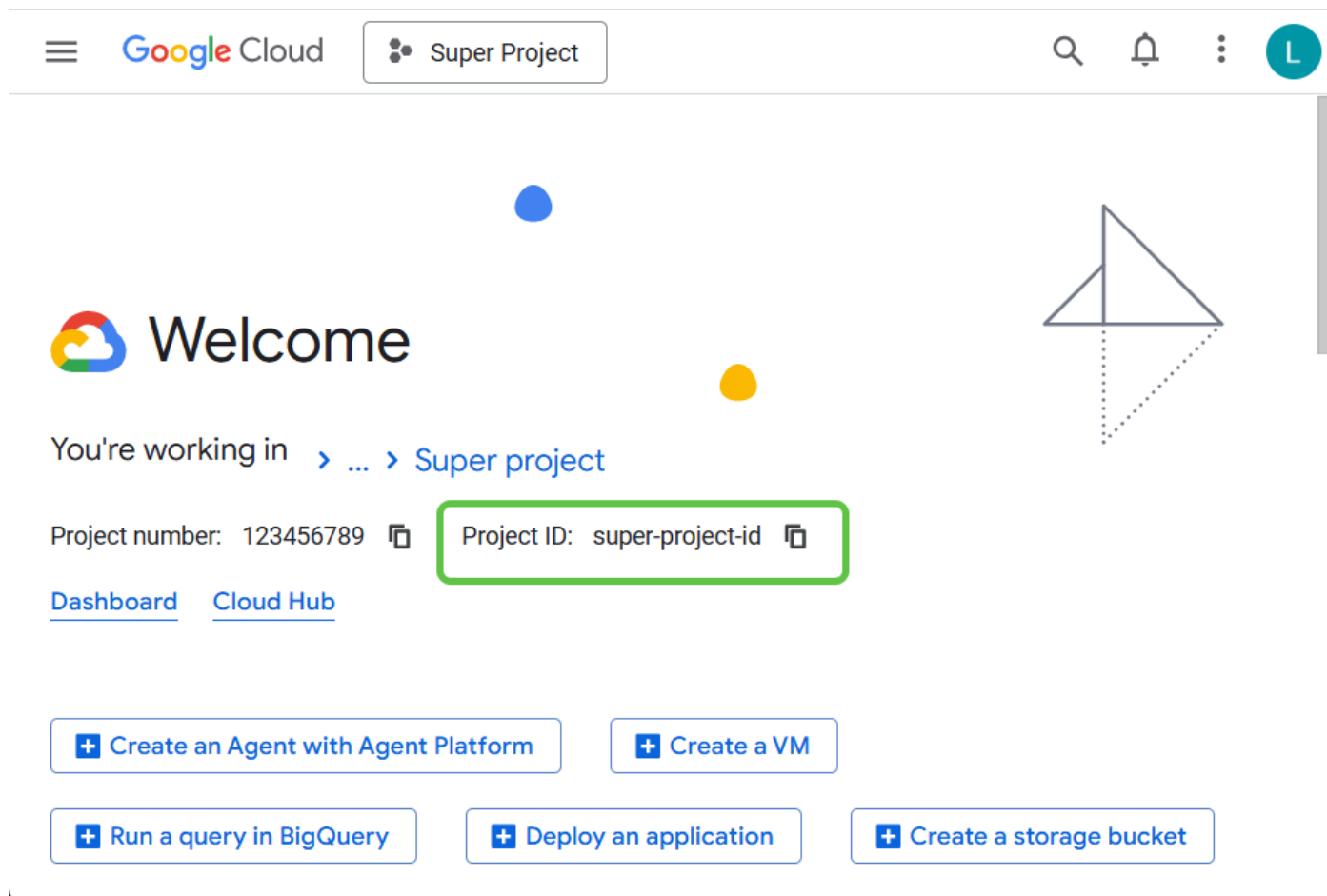
費用預估：本程式碼研究室建立的資源費用應低於 \$5 美元。

2. 事前準備 / 設定

建立或選取 Google Cloud 專案

您必須擁有已啟用計費功能的 Google Cloud 專案，才能使用本實驗室所需的服務。

1. 在 [Google Cloud 控制台](#) 的專案選取器頁面中，選取或建立 Google Cloud 專案。
2. 確認 Cloud 專案已啟用計費功能。瞭解如何[確認是否已啟用計費功能](#)。
3. 在 [Cloud 控制台首頁](#) 中找出專案 ID



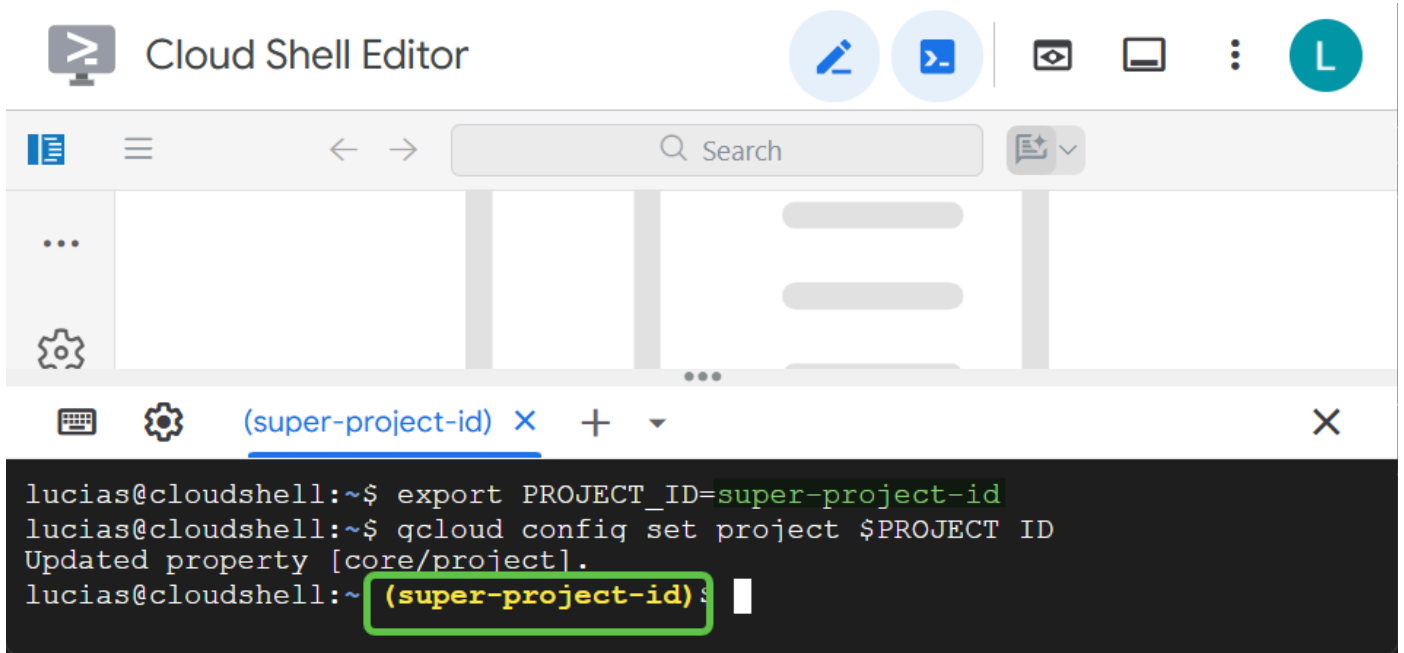
啟動 Cloud Shell

您將使用 [Google Cloud Shell](#) 做為執行環境。Cloud Shell 已預先安裝 `gcloud`、`git` 和其他必要工具。

1. 前往 [Google Cloud Shell](#)。
2. 如果出現提示訊息，請點選「授權」。
3. 在 Cloud Shell 終端機中設定環境變數，確保您在專案中工作：

將 `<YOUR_PROJECT_ID>` 替換為實際專案 ID。使用 Ctrl 或 Command + v 鍵貼到終端機。

```
export PROJECT_ID=<YOUR_PROJECT_ID>
gcloud config set project $PROJECT_ID
```



啟用必要的 API

執行下列指令，啟用 Spanner、BigQuery 和 Vertex AI 的 API：

```
gcloud services enable spanner.googleapis.com \
  bigquery.googleapis.com \
  aiplatform.googleapis.com \
  run.googleapis.com
```

複製存放區

複製包含應用程式程式碼和範例結構定義的存放區：

```
git clone https://github.com/GoogleCloudPlatform/cloud-spanner-samples.git
cd cloud-spanner-samples/spanner-bq-fraud-defense
```

3. 佈建基礎架構

您現在要在 BigQuery 中設定資料倉儲，並在 Spanner 中設定作業資料庫。

設定 BigQuery 資料集和連線

BigQuery 會分析遊戲的遙測串流。

在 Google Cloud Shell 的控制台中，執行下列指令，確認專案 ID 仍處於設定狀態：

將 <YOUR_PROJECT_ID> 替換為實際專案 ID。使用 Ctrl 或 Command + v 鍵貼到終端機。

```
export PROJECT_ID=<YOUR_PROJECT_ID>
gcloud config set project $PROJECT_ID
```

1. 建立 `game_analytics` BigQuery 資料集：

```
bq mk -d --location=US game_analytics
```

2. 建立連線，連結至儲存空間 bucket，並視需要連結至 Spanner：

```
bq mk --connection --location=US --project_id=$PROJECT_ID \
  --connection_type=CLOUD_RESOURCE unicorn-connection
```

3. 使用結構定義檔案，為 `GameplayTelemetry`、`AccountSignals`、`Players` 和 `ChatLogs` 資料表建立結構定義：

```
bq query --use_legacy_sql=false < bq_schema.sql
```

設定 Spanner

Google Cloud Spanner 會處理即時營運交易。在本實驗室中，我們會使用經濟實惠的 100 個處理單元 (PU) 執行個體。

1. 建立 Spanner 執行個體：

```
gcloud spanner instances create game-instance \
  --config=regional-us-central1 \
  --description="Game Instance" \
  --processing-units=100 \
  --edition=ENTERPRISE
```

2. 建立 Spanner 資料庫 `game-db`：

```
gcloud spanner databases create game-db --instance=game-instance
```

3. 使用應用程式資料表 (`Players` 、 `Transactions` 、 `AccountSignals`) 和屬性圖形 (`PlayerNetwork`) 更新 `Spanner` 結構定義：

```
gcloud spanner databases ddl update game-db --instance=game-instance --ddl-  
file=spanner_schema.sql
```

4. 為多模態向量搜尋建立 `AvatarSearchIndex`：

```
gcloud spanner databases ddl update game-db --instance=game-instance \  
--ddl="CREATE SEARCH INDEX AvatarSearchIndex ON Players(AvatarDescriptionTokens)"
```

將資料匯入 BigQuery

1. 將範例資料匯入 BigQuery：

```
bq load --source_format=AVRO game_analytics.GameplayTelemetry gs://sample-data-and-  
media/spanner-bq-fraud-heist/GameplayTelemetry  
bq load --source_format=AVRO game_analytics.AccountSignals gs://sample-data-and-  
media/spanner-bq-fraud-heist/AccountSignals  
bq load --source_format=AVRO game_analytics.Players gs://sample-data-and-media/spanner-bq-  
fraud-heist/Players  
bq load --source_format=AVRO game_analytics.ChatLogs gs://sample-data-and-media/spanner-bq-  
fraud-heist/ChatLogs
```

3. 前往 [BigQuery 控制台](#)，探索 `game_analytics` 資料集中的資料。

The screenshot displays the Google Cloud BigQuery console interface. On the left, a sidebar shows the project hierarchy: `super-project-id / Datasets / game_analytics / Tables / AccountSignals`. The main panel shows the `AccountSignals` table details. A notification indicates it is a partitioned table. Below the notification, tabs for `Schema`, `Details`, `Preview`, `Table Explorer`, `Preview`, `Insights`, `Lineage`, and `Data` are visible. The `Schema` tab is active, showing a table with the following fields:

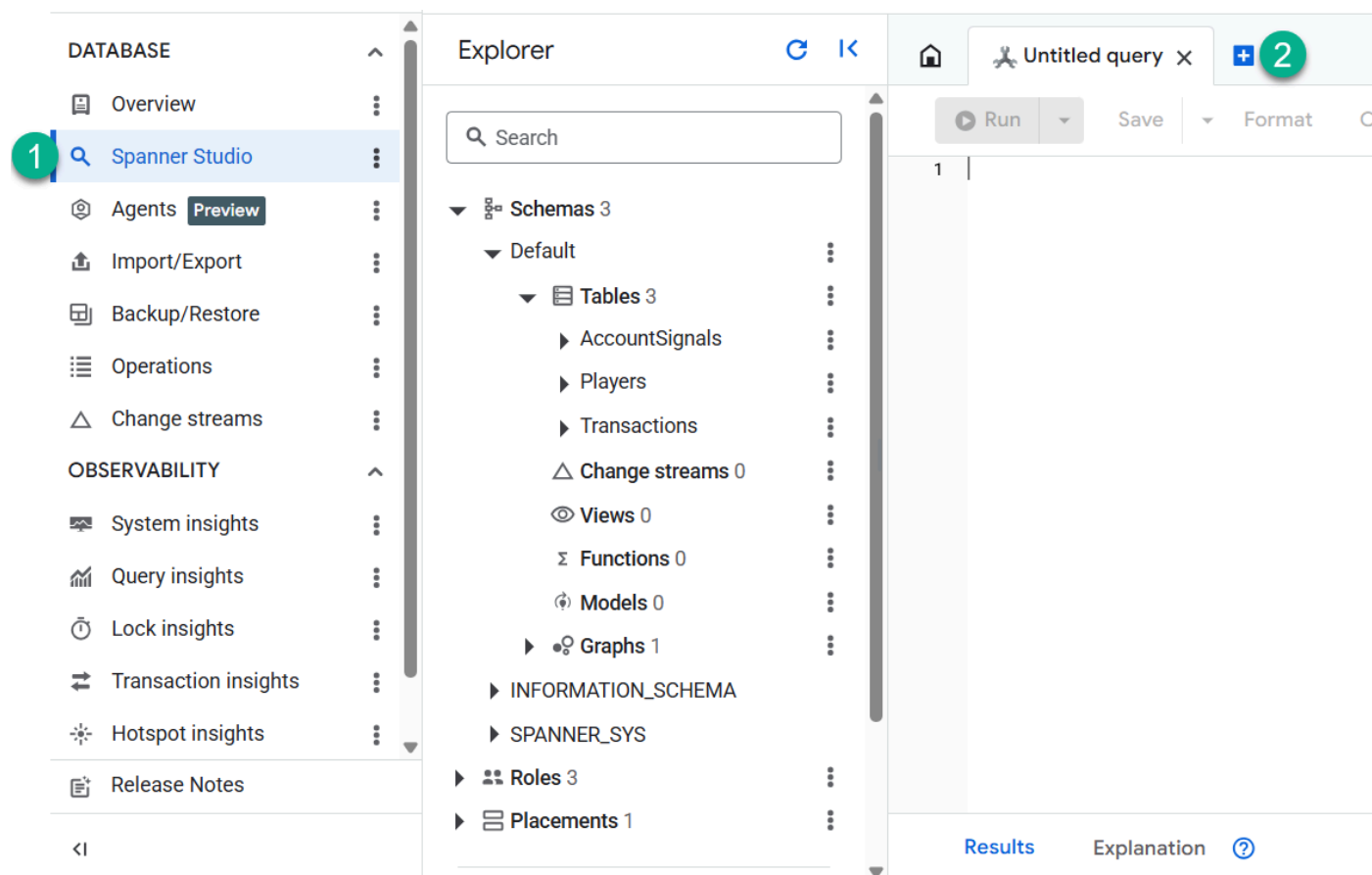
Field name	Type	Mode	Description	Key	Collation
SignalId	STRING	REQUIRED	-	-	-
PlayerId	STRING	REQUIRED	-	-	-
AlertType	STRING	REQUIRED	-	-	-
EventTime	TIMESTAMP	REQUIRED	-	-	-

At the bottom of the console, there are three buttons: `Edit schema`, `View row access policies`, and `Describe data`.

將資料匯入 Spanner

前往 [Spanner 控制台](#)，並探索 `game-db` 資料庫中的資料。

點選「Spanner Studio」，然後開啟「New Query (+)」。



將下列 `INSERT` 陳述式貼到查詢編輯器，然後點選「執行」：


```

-- Table: Players
INSERT INTO Players (
  PlayerId,
  Name,
  Species,
  Clan,
  AvatarDescription,
  ProfilePictureUrl,
  CreatedAt
) VALUES
  ('dc8cf07a-ac0f-48da-9f64-4f379492ble7', 'Pixel', 'Cat', 'CatClan', 'A heroic cat wearing a green tunic and a feathered cap', 'gs://sample-data-and-media/pixel_profile_booth.png', '2026-03-02T05:11:28.077335+00:00'),
  ('e82df4fb-0b6d-44dc-8609-70b41430af38', 'Rocky_1', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.077374+00:00'),
  ('ea3afac7-54f0-4f68-8ed5-a5b6bd386c59', 'Whiskers_2', 'Cat', 'CatClan', 'A sneaky black cat hiding in the shadows', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.077389+00:00'),
  ('f37d558b-fd0a-404c-a193-bc3a8a2edfba', 'Felix_3', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Abyssinian_1.jpg', '2026-03-02T05:11:28.077407+00:00'),
  ('f3687206-405e-43b6-afb0-8ca73eee5dd1', 'Luna_4', 'Cat', 'CatClan', 'A tabby cat with a red bandana', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.077419+00:00'),
  ('82383e2d-d3a2-481d-b0cf-bfe165ed9bfd', 'Luna_5', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Abyssinian_114.jpg', '2026-03-02T05:11:28.077429+00:00'),
  ('755c7aff-e538-4681-9b90-4b870a42ac72', 'Buddy_6', 'Dog', 'DogClan', 'A tough bulldog with a spiked collar', 'gs://sample-data-and-media/yorkshire_terrier_101.jpg', '2026-03-02T05:11:28.077439+00:00'),
  ('8a034e84-26b3-4198-8ec9-3749b1f60537', 'Charlie_7', 'Dog', 'DogClan', 'A police german shepherd with a badge', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.077462+00:00'),
  ('ca288a07-2bf8-46fa-a121-9bd0d0f44c64', 'Rocky_8', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/yorkshire_terrier_101.jpg', '2026-03-02T05:11:28.077474+00:00'),
  ('7b2881f0-289b-4ea4-9c0e-c1748249b70a', 'Bella_9', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/yorkshire_terrier_101.jpg', '2026-03-02T05:11:28.077484+00:00'),
  ('153f4022-a4ce-404a-8544-25d004fd34ad', 'Simba_10', 'Cat', 'CatClan', 'A tabby cat with a red bandana', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.077494+00:00'),
  ('3fb82b8e-75a1-49fd-8691-a9dabb42bc4b', 'Charlie_11', 'Dog', 'DogClan', 'A police german shepherd with a badge', 'gs://sample-data-and-media/staffordshire_bull_terrier_116.jpg', '2026-03-02T05:11:28.077507+00:00'),
  ('e37b6dcf-7ccb-47d0-8b9d-0da2fa30ad09', 'Felix_12', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077516+00:00'),
  ('0b395a7b-0673-4348-afd0-7cea9252629f', 'Bella_13', 'Dog', 'DogClan', 'A police german shepherd with a badge', 'gs://sample-data-and-media/yorkshire_terrier_101.jpg', '2026-03-02T05:11:28.077527+00:00'),

```

('64921a63-bea6-4a5c-9e49-7a817678c94f', 'Charlie_14', 'Dog', 'DogClan', 'A tough bulldog with a spiked collar', 'gs://sample-data-and-media/yorkshire_terrier_101.jpg', '2026-03-02T05:11:28.077537+00:00'),

('0d9040df-34a3-4e54-b660-a85e3c60a6fe', 'Felix_15', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Abyssinian_114.jpg', '2026-03-02T05:11:28.077546+00:00'),

('188c23a6-4c3b-4c25-9b40-9ec8ad7712a3', 'Max_16', 'Dog', 'DogClan', 'A fast greyhound wearing a racing vest', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.077556+00:00'),

('a2eae9c9-dbff-4704-b908-74518d687e17', 'Rocky_17', 'Dog', 'DogClan', 'A tough bulldog with a spiked collar', 'gs://sample-data-and-media/shiba_inu_105.jpg', '2026-03-02T05:11:28.077565+00:00'),

('b59d9b9a-a169-43c1-832d-7e7d4acfa19c', 'Felix_18', 'Cat', 'CatClan', 'A sneaky black cat hiding in the shadows', 'gs://sample-data-and-media/Abyssinian_1.jpg', '2026-03-02T05:11:28.077581+00:00'),

('7019b91a-d908-4c93-8ed2-d1c60e518630', 'Nala_19', 'Cat', 'CatClan', 'A royal siamese cat wearing a crown', 'gs://sample-data-and-media/Abyssinian_1.jpg', '2026-03-02T05:11:28.077595+00:00'),

('1a972a71-a530-407a-b3f7-1dd9b2532cec', 'Luna_20', 'Cat', 'CatClan', 'A sneaky black cat hiding in the shadows', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077604+00:00'),

('3bd0489d-3f43-428d-be2d-0434f0e04ed8', 'Luna_21', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.077613+00:00'),

('28d7ab21-7ca7-435b-bdc4-5266a381ef89', 'Whiskers_22', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.077623+00:00'),

('87d880d4-c949-40e3-8d76-9307b45d57fe', 'Rocky_23', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.077632+00:00'),

('b49a4523-83ff-4f59-8fca-54356928a18b', 'Nala_24', 'Cat', 'CatClan', 'A tabby cat with a red bandana', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077641+00:00'),

('1e049026-1231-4763-bc50-82cb51950fa2', 'Nala_25', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Abyssinian_1.jpg', '2026-03-02T05:11:28.077650+00:00'),

('e83caccd-e81e-454a-bf92-9d9a5a509e25', 'Simba_26', 'Cat', 'CatClan', 'A royal siamese cat wearing a crown', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077659+00:00'),

('45650b49-9701-4415-92a9-79073243d198', 'Simba_27', 'Cat', 'CatClan', 'A royal siamese cat wearing a crown', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.077668+00:00'),

('2eb95b36-c9c3-4720-849b-fc240c9434da', 'Bella_28', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/samoyed_97.jpg', '2026-03-02T05:11:28.077677+00:00'),

('491943d2-8aa3-4484-9d6d-afcab7ec579a', 'Buddy_29', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.077687+00:00'),

('004ac8b2-a37d-42c8-aba3-35b204f596f0', 'Max_30', 'Dog', 'DogClan', 'A police german shepherd with a badge', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.077697+00:00'),

('5cfla55e-7904-4937-a4bd-bf3d1fclc839', 'Whiskers_31', 'Cat', 'CatClan', 'A royal siamese

cat wearing a crown', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.077706+00:00'),

('4736fc20-1c22-49c2-af87-c35802302507', 'Luna_32', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.077715+00:00'),

('d98fc7ff-dd92-4723-994c-6267ef951bcc', 'Buddy_33', 'Dog', 'DogClan', 'A police german shepherd with a badge', 'gs://sample-data-and-media/staffordshire_bull_terrier_116.jpg', '2026-03-02T05:11:28.077797+00:00'),

('3de80488-e1b6-4908-a6ec-9c51f46f43b9', 'Luna_34', 'Cat', 'CatClan', 'A sneaky black cat hiding in the shadows', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.077821+00:00'),

('710128e0-4e1d-479c-acbc-497bbc5bc802', 'Bella_35', 'Dog', 'DogClan', 'A tough bulldog with a spiked collar', 'gs://sample-data-and-media/scottish_terrier_108.jpg', '2026-03-02T05:11:28.077836+00:00'),

('7e5d416f-f42c-410e-8e2a-19be0cdfc1e5', 'Simba_36', 'Cat', 'CatClan', 'A royal siamese cat wearing a crown', 'gs://sample-data-and-media/Abyssinian_114.jpg', '2026-03-02T05:11:28.077850+00:00'),

('b5af580c-4998-4f68-b52f-5c5b6a9d0aab', 'Buddy_37', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/scottish_terrier_108.jpg', '2026-03-02T05:11:28.077864+00:00'),

('145cc805-810a-4320-864d-ba6b5c6fbc33', 'Max_38', 'Dog', 'DogClan', 'A tough bulldog with a spiked collar', 'gs://sample-data-and-media/staffordshire_bull_terrier_116.jpg', '2026-03-02T05:11:28.077884+00:00'),

('34fa84c8-3356-4ed6-9ac5-967cd2baabac', 'Whiskers_39', 'Cat', 'CatClan', 'A royal siamese cat wearing a crown', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077894+00:00'),

('a50c967b-2e89-448d-be47-4df628c51572', 'Luna_40', 'Cat', 'CatClan', 'A royal siamese cat wearing a crown', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077906+00:00'),

('64aeeb22-42db-46bf-85bd-b21b135c9803', 'Simba_41', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.077915+00:00'),

('dec825d6-91f4-4f8f-b44b-74cf6b4b78f6', 'Felix_42', 'Cat', 'CatClan', 'A sneaky black cat hiding in the shadows', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.077924+00:00'),

('d9380e8d-8949-4849-af57-7a91a1a2b953', 'Bella_43', 'Dog', 'DogClan', 'A police german shepherd with a badge', 'gs://sample-data-and-media/staffordshire_bull_terrier_116.jpg', '2026-03-02T05:11:28.077933+00:00'),

('88a425a0-b499-4825-8a16-3d33b85feec9', 'Whiskers_44', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Abyssinian_114.jpg', '2026-03-02T05:11:28.077942+00:00'),

('e17ae312-37c6-4a66-9172-2d20d8528033', 'Simba_45', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Abyssinian_1.jpg', '2026-03-02T05:11:28.077958+00:00'),

('ae82282e-d380-4500-99e0-2dc4e276c0cf', 'Nala_46', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.077967+00:00'),

('9ce13d41-dede-4658-a7e4-7bc00372d65c', 'Simba_47', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077976+00:00'),

('8db26b5c-2c09-4578-89bf-f9afaf765e36', 'Whiskers_48', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-

02T05:11:28.077989+00:00'),
('eb113965-6ee6-459e-ba0f-fb6cb1d0ae34', 'Luna_49', 'Cat', 'CatClan', 'A tabby cat with a red bandana', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.078002+00:00'),
('3f064471-24a0-47cd-8927-925cb3e5df7f', 'Felix_50', 'Cat', 'CatClan', 'A tabby cat with a red bandana', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.078012+00:00'),
('ed20f879-2d42-4ba7-8df7-d696c4b6aafd', 'Whiskers_51', 'Cat', 'CatClan', 'A sneaky black cat hiding in the shadows', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.078021+00:00'),
('7e9bbec4-e2aa-4d67-b1c0-35f7db1d0dae', 'Buddy_52', 'Dog', 'DogClan', 'A fast greyhound wearing a racing vest', 'gs://sample-data-and-media/yorkshire_terrier_101.jpg', '2026-03-02T05:11:28.078030+00:00'),
('398de7e0-8f4f-44ba-adfe-503a157cfe7c', 'Rocky_53', 'Dog', 'DogClan', 'A fast greyhound wearing a racing vest', 'gs://sample-data-and-media/staffordshire_bull_terrier_116.jpg', '2026-03-02T05:11:28.078040+00:00'),
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('61dd4370-bff0-4d09-be31-19f19c4dad5d', 'Max_59', 'Dog', 'DogClan', 'A tough bulldog with a spiked collar', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.078099+00:00'),
('4e9dfce6-555b-434a-81b6-237c61b9b530', 'Merry_Cat_0', 'Cat', 'RobinHoods', 'A sneaky rebel cat wearing a green tunic and a feathered cap', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.078112+00:00'),
('686337b0-304e-4270-89c3-d015f9039294', 'Merry_Cat_1', 'Cat', 'RobinHoods', 'A sneaky rebel cat wearing a green tunic and a feathered cap', 'gs://sample-data-and-media/Abyssinian_114.jpg', '2026-03-02T05:11:28.078165+00:00'),
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('78d66ff4-0519-4157-9d95-76c2900ba7f9', 'Merry_Cat_4', 'Cat', 'RobinHoods', 'A sneaky rebel cat wearing a green tunic and a feathered cap', 'gs://sample-data-and-media/Abyssinian_114.jpg', '2026-03-02T05:11:28.078301+00:00'),
('822a8aae-57bd-460e-9643-69815990bec8', 'Merry_Cat_5', 'Cat', 'RobinHoods', 'A sneaky rebel cat wearing a green tunic and a feathered cap', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.078349+00:00'),

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('9f223b65-41a3-495e-940a-5a07d2ba4ba7', 'Merry_Cat_6', 'Cat', 'RobinHoods', 'A sneaky
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media/Abyssinian_114.jpg', '2026-03-02T05:11:28.078390+00:00'),
('3910b199-caf7-4822-8346-1ba9bd750c8e', 'Merry_Cat_7', 'Cat', 'RobinHoods', 'A sneaky
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media/Bengal_105.jpg', '2026-03-02T05:11:28.078427+00:00'),
('78c129b6-21e5-40c6-b9cf-d25ad5193e68', 'Merry_Cat_8', 'Cat', 'RobinHoods', 'A sneaky
rebel cat wearing a green tunic and a feathered cap', 'gs://sample-data-and-
media/Bengal_100.jpg', '2026-03-02T05:11:28.078463+00:00'),
('eba84e9d-bd10-466e-8fcd-899c0a868149', 'Merry_Cat_9', 'Cat', 'RobinHoods', 'A sneaky
rebel cat wearing a green tunic and a feathered cap', 'gs://sample-data-and-
media/Bombay_104.jpg', '2026-03-02T05:11:28.078508+00:00'),
('37a1e15c-a38c-4763-a2bd-042741bce012', 'Merry_Cat_10', 'Cat', 'RobinHoods', 'A sneaky
rebel cat wearing a green tunic and a feathered cap', 'gs://sample-data-and-
media/Bombay_104.jpg', '2026-03-02T05:11:28.078544+00:00'),
('c5337857-3b58-4b0c-8814-4585dcbf765f', 'Merry_Cat_11', 'Cat', 'RobinHoods', 'A sneaky
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media/Bombay_104.jpg', '2026-03-02T05:11:28.078587+00:00');

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-- Table: Transactions

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  ReceiverId,
  Amount,
  Timestamp,
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-- Table: AccountSignals

-- This is pushed by BigQuery through a continuous query if it is configured

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INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('50b64a6a-2e8f-4a0b-9742-c7e180949e82', '4e9dfce6-555b-434a-81b6-237c61b9b530', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078156+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('d2659e80-bce0-46b0-9570-6bb8b3c99d34', '686337b0-304e-4270-89c3-d015f9039294', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078198+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('10504018-b7a4-47a4-89b2-950fa492bbd4', '1627bc18-c42e-4599-b5f1-6f3d52669edb', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078244+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('ccbba78a-0f45-49b9-a58b-21777de250cf', '215b3d04-402a-4ac2-83ed-1edb9a421691', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078293+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('affa841d-0f9a-4b0f-ad6d-eb3e180263fe', '78d66ff4-0519-4157-9d95-76c2900ba7f9', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078337+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('1fbdd097-b431-4a21-9a5c-alf53db6d754', '822a8aae-57bd-460e-9643-69815990bec8', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078383+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('0fc87891-dd4b-494b-8c47-f6ccc2c592c7', '9f223b65-41a3-495e-940a-5a07d2ba4ba7', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078420+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('cc117869-7a6b-4e66-89bf-e3148ac8c6ea', '3910b199-caf7-4822-8346-1ba9bd750c8e', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078456+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('c85a78c6-3cc8-4262-8fd2-2fb9661a1b96', '78c129b6-21e5-40c6-b9cf-d25ad5193e68', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078499+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('ccdb20f8-cd05-42ae-b942-cb41396ec27d', 'eba84e9d-bd10-466e-8fcd-899c0a868149', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078537+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('721511f1-ebc4-45dc-adb5-9dafc5426bc3', '37ale15c-a38c-4763-a2bd-042741bce012', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078579+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('4d8b0f0d-211a-4537-8c79-c2fd815812f3', 'c5337857-3b58-4b0c-8814-4585dcbf765f', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078627+00:00');
```

這項作業會在幾分鐘內完成。完成後即可預覽資料。

All instances

Instance

game-instance: Overview

Google Standard SQL Database

game-db: Spanner Studio

Open in Dataplex

Learn

Explorer

Untitled query

Open CLI

Gemini settings

View in BigQuery

Valid

Search

Schemas 3

Default

Tables 3

AccountSignals

Columns 4

Keys 2

Indexes 0

Players

Transactions

Change streams 0

Views 0

Functions 0

Models 0

Graphs 1

INFORMATION_SCHEMA

Run

Save

Format

Clear

Documentation

1

2

3

4

5

6

SELECT

*

FROM

'AccountSignals'

LIMIT

10;

Preview data

Insert data

Drop table

Equivalent DDL

Open in Dataplex

Explanation

Export

Copy

Download

	PlayerId	AlertType	EventTime
	"b-494b-8c47-f6ccc2c592c7"	"9f223b65-41a3-495e-940a-5a07d2ba4ba7"	"SUSPICIOUS_MOVEMENT"
	"4-47a4-89b2-950fa492bbd4"	"1627bc18-c42e-4599-b5f1-6f3d52669edb"	"SUSPICIOUS_MOVEMENT"
	"1-4a21-9a5c-a1f53db6d754"	"822a8aae-57bd-460e-9643-69815990bec8"	"SUSPICIOUS_MOVEMENT"
	"4d8b0f0d-211a-4537-8c79-c2fd815812f3"	"c5337857-3b58-4b0c-8814-4585dcbf765f"	"SUSPICIOUS_MOVEMENT"
	"50b64a6a-2e8f-4a0b-9742-c7e180949e82"	"4e9dfce6-555b-434a-81b6-237c61b9b530"	"SUSPICIOUS_MOVEMENT"
	"721511f1-ebc4-45dc-adb5-9dafc5426bc3"	"37a1e15c-a38c-4763-a2bd-042741bce012"	"SUSPICIOUS_MOVEMENT"
	"affa841d-0f9a-4b0f-ad6d-eb3e180263fe"	"78d66ff4-0519-4157-9d95-76c2900ba7f9"	"SUSPICIOUS_MOVEMENT"
	"c85a78c6-3cc8-4262-8fd2-2fb9661a1b96"	"78c129b6-21e5-40c6-b9cf-d25ad5193e68"	"SUSPICIOUS_MOVEMENT"

步驟 4 · 約 1 分鐘

4. Watchdog (BigQuery 持續查詢和 Spanner 同步)

我們的第一道防線是串流至 BigQuery 的遙測資料。我們想監控可疑移動 (例如不可能的距離)，並即時將快訊推送回 Spanner。

在實際情況中，您會使用 [BigQuery 持續查詢](#) 和反向 ETL 串流這項資料。不過，這需要 [預訂 ENTERPRISE 以上版本](#)。

如果預訂時間可供預約，指令會如下所示。如果您尚未設定預訂，則不需要將這項資訊複製到控制台中：

```
EXPORT DATA
  OPTIONS (
    uri = 'https://spanner.googleapis.com/projects/<YOUR_PROJECT_ID>/instances/game-
instance/databases/game-db',
    format='CLOUD_SPANNER',
    spanner_options="{ \"table\": \"AccountSignals\" }"
  ) AS
SELECT
  GENERATE_UUID() as SignalId,
  PlayerId,
  'SUSPICIOUS_MOVEMENT' as AlertType,
  CURRENT_TIMESTAMP() as EventTime
FROM `game_analytics.GameplayTelemetry`
WHERE
  EventType = 'player_move'
  AND (LocationX > 1000 OR LocationY > 1000);
```

如要建立持續查詢，請在 BigQuery 控制台的 SQL 工作區中，依序點選「更多」>「建立持續查詢」。

這項查詢會做為反向 ETL 引擎，確保交易系統 (Spanner) 能立即得知分析系統 (BigQuery) 偵測到的異常狀況。

在本實驗室中，我們在 Spanner 中人為插入了一些交易。

步驟 5 · 約 10 分鐘

5. 多模態偵探 (Spanner Graph 和向量搜尋)

現在 Spanner 具有「高風險」信號，您可以調查詐欺集團。我們會使用 **Spanner Graph** 呈現金融網路，並找出主謀。

在 **Spanner Studio** 中執行這些查詢

圖表：找出幕後主使者

這項查詢會追蹤金融交易網路，受害者會將款項轉給竊賊，竊賊再轉給主節點。並依首領分組，加總戰利品。

複製到 Spanner Studio，然後點選「Run」（執行）。

```
GRAPH PlayerNetwork
MATCH (victim)-[:Transfers]->(thief)-[t:Transfers]->(boss)
RETURN
  boss.Name AS RingLeader, COUNT(t) AS TributesReceived,
  SUM(t.Amount) AS TotalLoot
GROUP BY RingLeader
ORDER BY TotalLoot DESC
LIMIT 5;
```

「Pixel」應會顯示為獲得最多致敬的收件者。

多信號調查

現在，我們將 Graph 結果與先前從 BigQuery 推送的即時行為信號合併。我們想找出將款項匯給「Pixel」且因可疑活動而遭到標記的玩家。

```
SELECT DISTINCT
  p.Name,
  s.AlertType as BQ_Signal,
  s.EventTime as SignalTime
FROM GRAPH_TABLE (
  PlayerNetwork
  MATCH (associate:Players)-[:Transfers]->(boss:Players)
  WHERE boss.Name = 'Pixel'
  RETURN DISTINCT associate.Name
) as g
JOIN Players p
  ON p.Name = g.Name
JOIN AccountSignals s
  ON p.PlayerId = s.PlayerId
ORDER BY s.EventTime DESC;
```

向量搜尋：找出機器人帳戶

這些是真人玩家還是有組織的機器人集團？使用**向量搜尋**功能，找出個人資料說明與「Pixel」極為相似的帳戶。

```
SELECT
  Name, AvatarDescription,
  COSINE_DISTANCE(AvatarEmbedding, (SELECT AvatarEmbedding FROM Players WHERE Name =
'Pixel')) as Similarity
FROM Players
WHERE Name != 'Pixel'
ORDER BY Similarity ASC
LIMIT 5;
```

相似度分數越低，表示越接近「Pixel」的向量。如果說明類似，很可能就是機器人！

我們也可以在 `MATCH` 子句中套用純量函式：

```
GRAPH PlayerNetwork
MATCH (associate:Players)-[:Transfers]->(boss:Players)
WHERE boss.Name = 'Pixel'
ORDER BY (
  COSINE_DISTANCE(associate.AvatarEmbedding, (SELECT AvatarEmbedding FROM Players WHERE Name
= 'Pixel'))
) ASC
RETURN DISTINCT associate.Name
```

步驟 6 · 約 8 分鐘

6. 揭開劇情 (BigQuery 屬性圖和 GCS 整合)

我們已逮捕首腦，但需要瞭解他們如何協調這場大規模的「鮪魚搶劫案」。我們將使用 **BigQuery 屬性圖** 查詢遊戲聊天記錄，追蹤 BigQuery 中的通訊模式。

在 **BigQuery Studio** 中執行下列查詢：

BigQuery 屬性圖

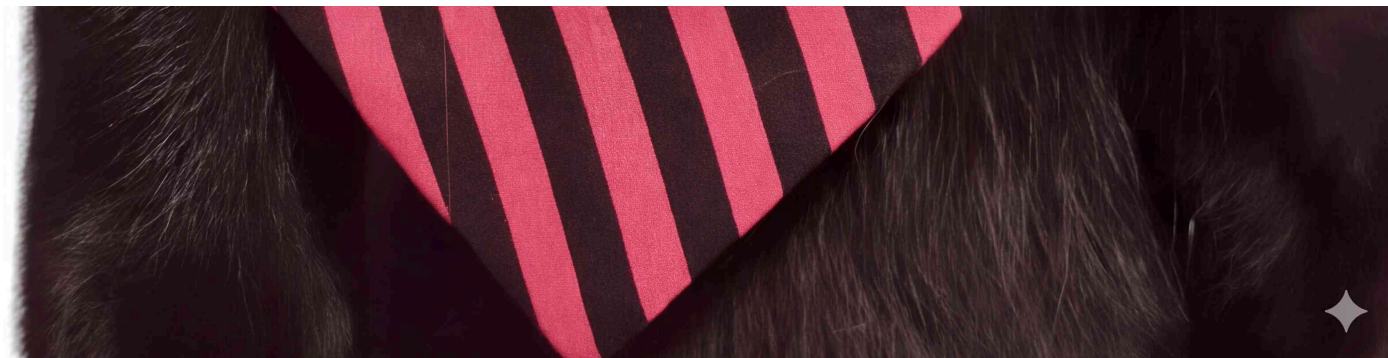
追蹤「Pixel」和其他播放器之間的通訊：

```
GRAPH game_analytics.CatChatNetwork
MATCH (p1:Players)-[c:Communicates]->(p2:Players)
WHERE p1.Name = 'Pixel' OR p2.Name = 'Pixel'
RETURN
  p1.Name AS Sender,
  p2.Name AS Receiver,
  c.Message,
  -- Resolving structured metadata from ObjectRef
  p1.ProfilePictureUrl.uri AS SenderProfilePic
ORDER BY Message DESC;
```

觀察「Operation Fishbowl is a go」（魚缸行動開始）和「Diverting funds to the central tuna reserve」（將資金轉移至中央鮪魚儲備）等訊息。您可以瞭解如何使用 BigQuery 屬性圖表，分析以非結構化資料 (使用 `ProfilePictureUrl.uri` 的 GCS 圖片參照) 擴增的通訊內容。

如果按照結果中的 **GCS 連結** 操作，您會看到播放器的圖片：





這項分析查詢會進一步比較詐騙集團成員之間的即時通訊模式和圖片。

如要執行這項操作，請先宣告多模態模型，為儲存在 Cloud Storage bucket 中的個人資料相片生成嵌入內容。這個模型會透過初始設定中建立的連線進行連線，因此您也會將 Vertex AI 使用者權限授予附加至該連線的技術使用者。

將「PROJECT_ID」換成您的專案 ID。

```
GRANT `roles/aiplatform.user`  
ON PROJECT `<<PROJECT_ID>>`  
TO "connection:<<PROJECT_ID>>.us.unicorn-connection";
```

現在可以建立連線了。

```
CREATE OR REPLACE MODEL `game_analytics.multimodal_model`  
  REMOTE WITH CONNECTION `us.unicorn-connection`  
  OPTIONS (ENDPOINT = 'multimodalembembedding@001');
```

如果因權限錯誤而失敗 (即「The bqcx-12345745345345@gcp-sa-bigquery-condel.iam.gserviceaccount.com does not have the permission to access or use the endpoint...」)，請稍候幾分鐘，讓權限生效，然後再試一次。

下列查詢會使用 AI.GENERATE_EMBEDDING 函式，處理儲存空間 bucket 中的圖片並建立嵌入。接著使用 COSINE_DISTANCE 比較這些嵌入內容，深入瞭解對話記錄和個人資料相片的相似程度。

```

-- BigQuery Property Graph: Tracing communication patterns in chat logs
-- AND calculating distance between auto-embedded chat message and profile picture
-- BigQuery Property Graph: Tracing communication patterns
-- AND identifying similarity AMONG the fraudsters themselves
WITH GraphResults AS (
  SELECT *
  FROM GRAPH_TABLE(
    game_analytics.CatChatNetwork
    MATCH (p1:Players)-[c:Communicates]->(p2:Players)
    WHERE p1.Name = 'Pixel' OR p2.Name = 'Pixel'
    RETURN
      p1.Name AS Sender,
      c.Message,
      p1.ProfilePictureUri.uri AS SenderProfilePic,
      c.MessageEmbedding.result AS MessageEmbedding
  )
),
UniquePics AS (
  SELECT DISTINCT SenderProfilePic AS uri FROM GraphResults
),
PicEmbeddings AS (
  SELECT embedding, uri
  FROM AI.GENERATE_EMBEDDING(
    MODEL game_analytics.multimodal_model,
    (
      SELECT OBJ.MAKE_REF(uri, 'us.unicorn-connection') as content, uri
      FROM UniquePics
    )
  )
),
CatData AS (
  -- Distinct list of players (excluding Pixel) with their embeddings and HTTPS Pic URLs
  SELECT DISTINCT
    g.Sender,
    g.MessageEmbedding,
    g.Message,
    p.embedding AS PicEmbedding,
    REPLACE( g.SenderProfilePic, 'gs://sample-data-and-media/spanner-bq-fraud-
    heist/profile_pics/', 'https://storage.mtls.cloud.google.com/sample-data-and-media/spanner-
    bq-fraud-heist/profile_pics/') AS SenderProfilePic
  FROM GraphResults g
  LEFT JOIN PicEmbeddings p ON g.SenderProfilePic = p.uri
  WHERE g.Sender != 'Pixel'
    AND g.MessageEmbedding IS NOT NULL
    AND p.embedding IS NOT NULL
)
SELECT
  c1.Sender AS Fraudster_A,
  c2.Sender AS Fraudster_B,
  c1.SenderProfilePic AS Pic_A,
  c2.SenderProfilePic AS Pic_B,
  c1.Message,

```

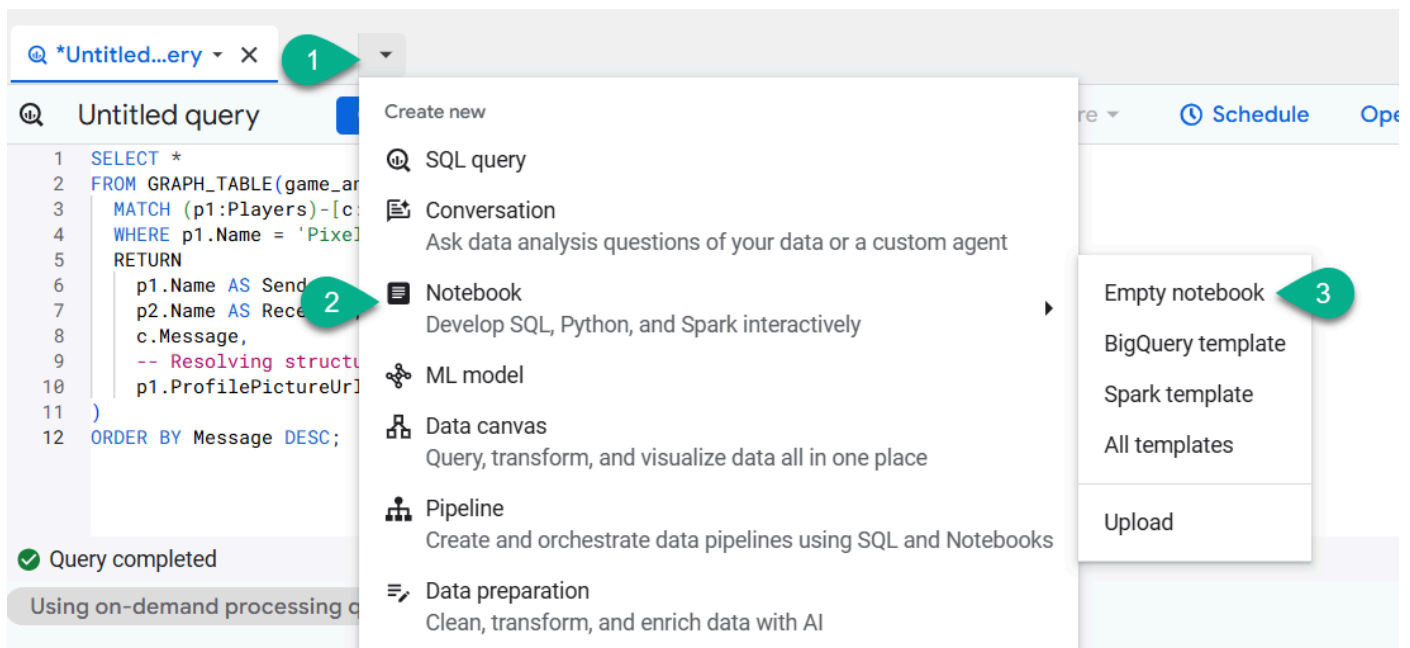
```
-- Compare chat messages between Fraudster A and B
COSINE_DISTANCE(c1.MessageEmbedding, c2.MessageEmbedding) AS MessageDistance,
-- Compare profile pictures between Fraudster A and B
COSINE_DISTANCE(c1.PicEmbedding, c2.PicEmbedding) AS PictureDistance
FROM CatData c1
CROSS JOIN CatData c2
WHERE c1.Sender < c2.Sender -- Avoid self-comparison and duplicate pairs (A-B and B-A)
      AND c1.SenderProfilePic <> c2.SenderProfilePic
ORDER BY PictureDistance ASC, MessageDistance ASC
LIMIT 10;
```

開啟個人資料相片後，你會發現氏族成員的呈現方式十分相似。

以視覺化方式呈現詐欺集團

您可以使用筆記本和 Python Cell Magic，將詐騙集團視覺化。這樣我們就能輕鬆查看圖表結果。如要進一步瞭解這項功能，請參閱[說明文件](#)中的視覺化內容。

在 BigQuery Studio 中，依序點按「更多」 > 「Notebook」 > 「空白 Notebook」。



如果是第一次使用筆記本，系統會要求您提供區域。本實驗室使用 `us`，因此請在 `us` 多區域中選擇一個區域。您可能也需要啟用 Notebooks API。

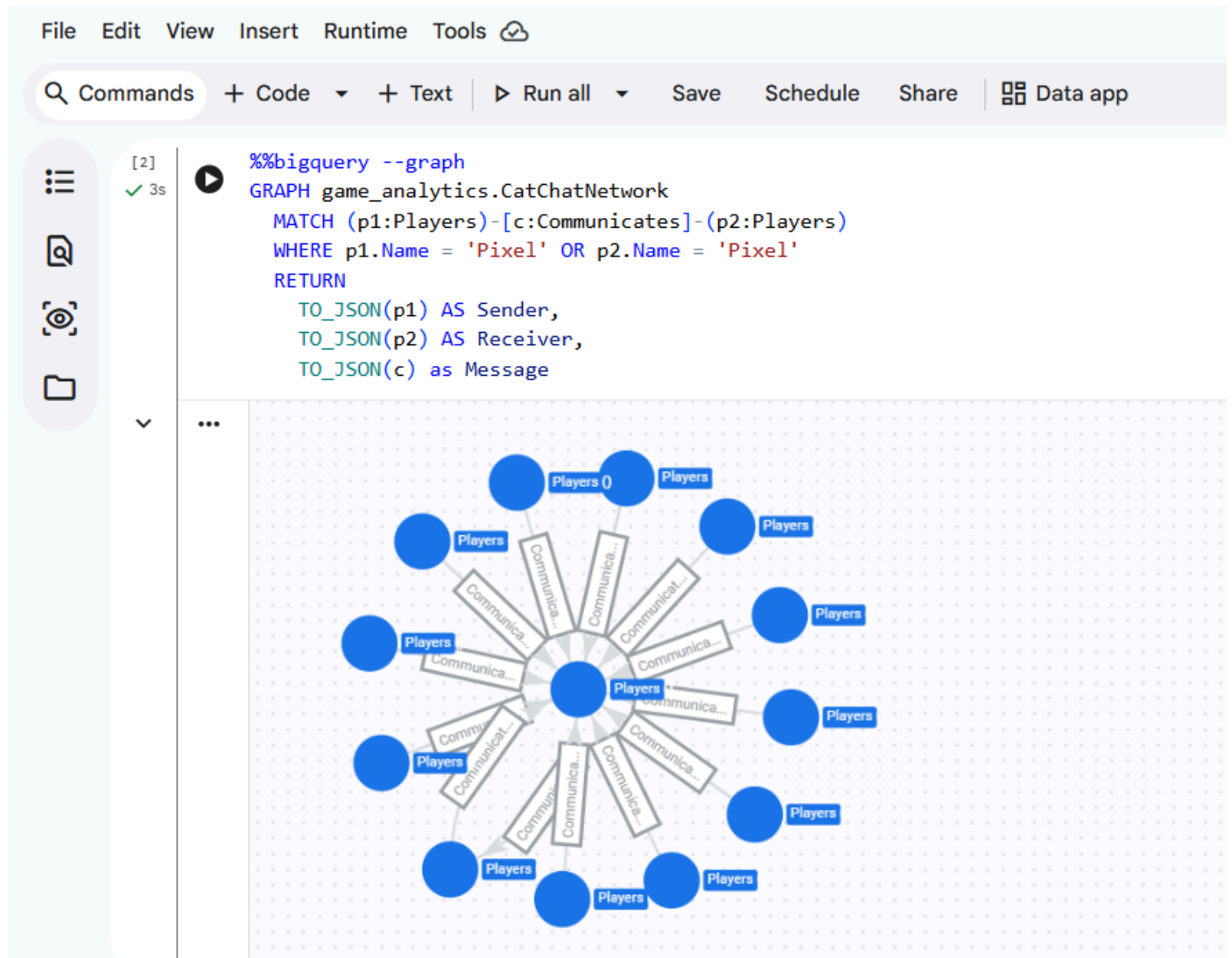
將下列內容貼到程式碼儲存格：

```
!pip install bigquery-magics==0.12.1
```

使用「+ Code」按鈕建立新的儲存格，並貼上下列內容：


```
%%bigquery --graph
GRAPH game_analytics.CatChatNetwork
MATCH p=(p1:Players)-[c:Communicates]-(p2:Players)
WHERE p1.Name = 'Pixel' OR p2.Name = 'Pixel'
RETURN TO_JSON(p) AS full_path
```

按一下「Run all」(全部執行)。大約一分鐘後，您應該會看到通訊網路的圖表。



步驟 7 · 約 5 分鐘

7. 清除

如要避免系統向您的 Google Cloud 帳戶收取本程式碼研究室所用資源的費用，請刪除您建立的資源。

刪除 Spanner 執行個體

```
gcloud spanner instances delete game-instance
```

刪除 BigQuery 資料集

```
bq rm -r -f -d game_analytics
```

或者，刪除專案

如果您是為了這個實驗室建立新專案，可以刪除整個專案：

```
gcloud projects delete <YOUR_PROJECT_ID>
```

步驟 8 · 約 1 分鐘

8. 恭喜！

恭喜！您已成功使用 Spanner 和 BigQuery 建構即時詐欺防禦機制。

您已學會如何：

- 使用 **BigQuery 持續查詢**，將即時洞察資料推送至 Spanner。
- 使用 **Spanner Graph** 追蹤財務關係。
- 使用 **Spanner 向量搜尋**，對非結構化資料執行相似度查詢。
- 使用 **BigQuery 圖**追蹤通訊網路。

後續步驟

- 參閱 [Spanner Graph 說明文件](#)。
 - 如需更多程式碼範例，請前往 [Spanner Graph GitHub 存放區](#)。
 - 進一步瞭解 [BigQuery Graph](#)。
 - 請參閱「[BigQuery 持續查詢](#)」。
-